How to Write a Science Research Report

A Research Report

A practical guide for school, college and university students: planning an investigation, recording honest results, and writing them up so that another person can understand, check and repeat the work.

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Abstract

A science research report is a written record of an investigation. It explains what question was asked, how the answer was sought, what was found, and what the findings mean. Unlike an essay, which mainly argues a point of view, a research report is judged by the quality of its evidence: whether the method was fair, the measurements were honest, and the conclusions follow from the results rather than from the writer's wishes.

This guide explains the report step by step. It begins with the purpose of a report and the people who read it, then describes the IMRaD structure used by most scientific journals and school and university courses: Introduction, Methods, Results and Discussion. Each section is explained in detail, with examples of good and weak writing, advice on titles, abstracts, tables and graphs, and guidance on ethics, safety and referencing.

The guide also contains a complete worked example. Chapter 10 presents a short model report based on a clearly labelled fictional experiment on the effect of light on plant growth. The example shows how the same principles apply from the title page to the final checklist, and it can be used as a template for a student's own first report.

Finally, the guide lists the most common mistakes made by new writers — overstating results, hiding unexpected findings, copying text without acknowledgement, and confusing a conclusion with a guess — and provides a final checklist that can be used before any report is submitted. A student who follows this guide should be able to produce a report that is clear, honest, complete and ready for a teacher, examiner or other reader to assess fairly.

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Chapter 1. Introduction: What a Science Research Report Is and Who Reads It

Science advances by sharing. An experiment that is never written down helps nobody except the person who performed it, and even that person will forget the details within weeks. A research report is the bridge between doing science and communicating science. It turns private notes, measurements and observations into a public document that other people can read, understand, criticise and build upon.

A research report is not a story about how hard the work was, and it is not an advertisement for the writer's cleverness. It is a factual, organised account. The reader should finish it knowing exactly what was investigated, how it was investigated, what happened, and what can reasonably be concluded. If any of those four things is missing, the report is incomplete, however well written the rest may be.

1.1 What Counts as a Research Report?

A research report describes original work: an experiment, a survey, an observation or a measurement that the writer (or the writer's group) actually carried out. It is different from a review, which summarises other people's work, and different from a textbook chapter, which explains established knowledge. A report may include background reading, but its centre is new evidence collected by the writer.

Reports vary in length. A school practical may need only two or three pages. A college project may need ten to twenty pages. A university dissertation may run to fifty pages or more, and a journal article is usually eight to fifteen pages. The length changes; the structure and the standards of honesty do not. This guide is written mainly for school and early university students, but its principles apply at every level.

1.2 Who Reads a Report, and What Do They Want?

Most reports have more than one reader, and each reader wants something slightly different. A teacher or examiner wants to see that the student understood the science, followed a fair method, recorded results honestly and thought carefully about what they mean. A classmate wants to understand the work without having been present in the laboratory. A future student may want to repeat the experiment, so the method must be complete enough to follow.

In professional science the readers include other researchers, journal editors and reviewers. They ask harder questions: Is the question worth asking? Was the method capable of answering it? Are the results strong enough to support the conclusion? Could another explanation fit the same data? A student report will not be judged as

harshly as a journal article, but writing as if a sceptical expert will read it is excellent training. It forces the writer to be precise, to give evidence for every claim, and to admit uncertainty where it exists.

1.3 The Qualities of a Good Report

- Clarity: short sentences, ordinary words where possible, and technical terms defined when they are first used.
- Accuracy: numbers copied correctly, units stated, and no claim made that the data do not support.
- Completeness: enough detail for another student to repeat the work and obtain similar results.
- Honesty: unexpected, negative and messy results reported alongside the neat ones.
- Organisation: a standard structure, clear headings, and a logical order that the reader can follow without effort.

These qualities matter more than elegant style. A plain report that is clear and honest will always score better than a beautifully written report that hides its method or exaggerates its findings.

1.4 How to Use This Guide

Read Chapters 1 to 3 before starting an investigation, because the title, the aim and the structure should be decided early, not added at the end. Write the Methods section while the experiment is still fresh, ideally on the same day the work is done. Write the Results as soon as the data are collected, before deciding what they mean. Write the Introduction and Discussion last in substance, even though the Introduction appears first in the finished report: it is much easier to introduce a study once you know what it actually found.

Golden rule
Write for a reader who was not there.
If a stranger cannot repeat your experiment from your Methods section alone, the section is not finished.

Chapter 2. The IMRaD Structure: Introduction, Methods, Results and Discussion

Most science reports follow the same four-part structure, known by the letters IMRaD: Introduction, Methods, Results and Discussion. The structure is used by school practicals, university projects and professional journals around the world. Learning it once means a student can read almost any scientific paper and can write a report that any science teacher will recognise.

2.1 What Each Part Does

The Introduction explains why the study was done. It gives the background knowledge the reader needs, identifies a gap or a problem, states the aim or research question, and often states a hypothesis — a testable prediction. It answers the question: why should anyone care?

The Methods section explains what was done. It lists the materials and equipment, describes the procedure step by step, explains how variables were controlled and measured, and notes any ethical or safety issues. It answers the question: how was the study done, in enough detail to repeat it?

The Results section presents what was found, without interpretation. It uses tables and graphs, states the main patterns, and gives the numbers, including averages and the range of values. It answers the question: what happened?

The Discussion explains what the results mean. It compares the findings with the hypothesis and with earlier work, explains unexpected results, admits limitations, and suggests what should be studied next. It answers the question: so what?

Around these four parts sit the supporting sections: a title, an abstract (a short summary of the whole report), a conclusion, a list of references, and appendices for extra material such as raw data or a full questionnaire.

2.2 The Order of Writing Is Not the Order of Reading

Beginners often try to write a report from the first page to the last. Experienced writers do not. A practical order is: Methods first (while the memory is fresh), then Results (tables and graphs from the raw data), then Discussion (what the results mean), then Introduction (now that the story is known), then Abstract and Title (last of all, because they summarise everything), and finally the Conclusion and References. The reader will never know the order in which the sections were written; they will only notice that each section is clear and consistent with the others.

Section	Main question	Write it...
Title	What was studied?	Last
Abstract	What happened, briefly?	Last
Introduction	Why was it done?	Late
Methods	How was it done?	First
Results	What was found?	Second
Discussion	What does it mean?	Third
Conclusion	What is the answer?	Near the end
References	Whose work was used?	As you go

Table 1. The parts of a research report, the question each one answers, and a sensible order for writing them.

2.3 One Idea Per Paragraph

A report is built from paragraphs, and each paragraph should contain one idea. A paragraph that tries to explain the background, describe the method and announce a result all at once will confuse the reader. Start a new paragraph when the topic changes, and begin each paragraph with a sentence that tells the reader what the paragraph is about. Short paragraphs are not a weakness in scientific writing; they are a kindness to the reader.

Chapter 3. Writing a Clear Title and Abstract

The title and the abstract are the most-read parts of any report. A teacher deciding which reports to read first, a student searching a library, or a researcher scanning a journal will all judge a report by these two short pieces. They deserve more care than their length suggests.

3.1 The Title

A good title names the topic and, where possible, the key variable or finding. It should be specific enough that a reader knows what was studied, but short enough to remember. Ten to fifteen words is a good target for a school or college report.

Weak title: "Plants." This says nothing. It does not say which plants, what was tested, or what was measured.

Better title: "The Effect of Light Intensity on the Growth of Bean Seedlings Over Four Weeks." The reader now knows the factor tested (light intensity), the organism (bean seedlings), the measurement (growth) and the time period (four weeks).

Avoid titles that promise more than the study delivers. A study of ten bean seedlings in a classroom cannot honestly be titled "How Light Controls All Plant Growth." Avoid jokes, puns and questions as titles in formal reports; a clear statement is stronger. Do not use abbreviations in a title unless they are extremely well known, and never put the writer's opinion in the title — the evidence belongs in the Results and Discussion.

3.2 The Abstract

The abstract is a summary of the whole report in one paragraph, usually 150 to 250 words. It is written after the rest of the report is finished, because it must summarise work that already exists. A complete abstract answers four questions in order: What was the aim? How was it done? What was found? What does it mean?

Write the abstract in the past tense, because the work is finished. Use numbers for the main results, not vague words: write "seedlings in bright light grew on average 4.2 cm taller" rather than "seedlings in bright light grew much taller." Do not include references, tables, graphs or abbreviations that the reader has not yet met. Do not add information that is not in the main report.

Example abstract (fictional study, 148 words)

"This study investigated whether light intensity affects the growth of bean seedlings. Thirty seedlings were divided into three groups and grown for four weeks under low, medium or high light, with water, soil and temperature kept the same. Height was measured every seven days. After four weeks, seedlings in high light averaged 24.6 cm tall, compared with 18.3 cm in medium light and 11.2 cm in low light. The differences were consistent across all measurement days. The results support the hypothesis that greater light intensity increases growth in bean seedlings within the range tested. Further work could test whether the effect continues at still higher light levels or whether very bright light damages the leaves."

Notice how this short paragraph contains the aim, the method, the main numbers and the conclusion. A reader who reads only the abstract still learns what the study found.

3.3 Keywords

Some reports ask for three to five keywords after the abstract. These are single words or short phrases that describe the topic — for example, "bean seedlings, light intensity, plant growth, photosynthesis." Keywords help other people find the report in a database. Choose words a reader would actually search for, not words that are already in the title alone.

Chapter 4. The Introduction: Background, Problem and Hypothesis

The Introduction is the first full chapter of the report, but it is often the hardest to write well. Its job is to take a reader who knows general science and bring them, step by step, to the exact question this study answers. A common shape is the funnel: start broad, narrow to the specific problem, and end with the aim and hypothesis.

4.1 Background: What Is Already Known?

Begin with the established science the reader needs. For a study of light and plant growth, the background would explain photosynthesis in a few sentences: plants use light energy to make food, and light intensity affects the rate of photosynthesis up to a point. Give only the background that is relevant. A history of plant science from ancient times is not needed, and padding the introduction with unrelated facts wastes the reader's time and suggests the writer does not know what matters.

Every factual claim that comes from another source should be credited. In a school

report this may be a simple reference to a textbook or a named source in the References list. In university work a formal citation style is required. The rule is the same at every level: if the idea is not yours and not common knowledge, say where it came from.

4.2 The Problem and the Gap

After the background, identify the problem. Perhaps earlier studies disagree, or a question has never been tested with this particular plant, or a classroom observation raised a question nobody has answered. The gap does not need to be world-changing. "We know that light affects growth, but we do not know how much difference light intensity makes for bean seedlings in their first four weeks" is a perfectly good gap for a school investigation. A clear, small gap is better than a grand, vague one.

4.3 Aim, Research Question and Hypothesis

The aim is a single sentence stating what the study sets out to do: "The aim of this study was to investigate the effect of light intensity on the growth of bean seedlings over four weeks." The research question turns the aim into a question: "Does light intensity affect the growth of bean seedlings?" The hypothesis is a testable prediction, usually written before the experiment begins: "Bean seedlings grown in high light will grow taller than seedlings grown in low light over four weeks."

A good hypothesis is specific and falsifiable — that is, it must be possible for the results to prove it wrong. "Light is good for plants" cannot be tested, because "good" is not measured. "Seedlings in high light will be taller" can be tested with a ruler. State the hypothesis clearly, and do not change it after seeing the results. If the results disagree with the hypothesis, that is a finding, not a failure, and it must be reported honestly in the Discussion.

4.4 Variables

Every experimental study should name its variables. The independent variable is the factor the investigator changes — in this example, light intensity. The dependent variable is the factor that is measured — seedling height. Controlled variables are the factors kept the same for all groups — water, soil, pot size, temperature and the type of seed. Naming the variables in the Introduction shows the reader that the study was designed before it was carried out, and it prepares them for the Methods chapter that follows.

Chapter 5. Methods: Materials, Procedure and Ethics

The Methods section is the most practical part of a report. Its test is simple: could another student, reading only this section, repeat the experiment and obtain similar results? If the answer is no, the section needs more detail. Vague methods are the most common weakness in student reports, and the easiest to fix.

5.1 Materials and Equipment

List everything that was used, with quantities and sizes where they matter. Include the organism or sample (for example, "thirty bean seeds of the same variety, bought from the same packet"), the equipment (pots, soil, ruler, lamp), and the measuring instruments with their precision ("a ruler marked in millimetres" or "a balance accurate to 0.1 g"). If a chemical was used, give its name and concentration. If software was used for analysis, name it.

Do not list everyday items in excessive detail — nobody needs the brand of the pencil used to label the pots — but do include anything that could affect the result. The size of the pots matters, because a small pot limits root growth. The type of soil matters. The distance of the lamp from the plants matters, because light intensity falls quickly with distance.

5.2 Procedure: Step by Step

Describe the procedure in the order it was carried out, in numbered steps or in clear paragraphs, using the past tense. Include the numbers that matter: how many plants, how much water, how often, for how long, and how measurements were taken. For example: "Each pot received 50 ml of water every second day. Height was measured from the soil surface to the tip of the tallest leaf, every seven days, using the same ruler."

Explain the design: how many groups were used, how many repeats were in each group, and how the groups differed. Repeats matter, because a single plant may grow badly for a random reason. Ten plants in each group give a much more reliable average than one plant in each group. Explain how fairness was protected: the pots were placed so that each group received the same temperature, the plants were rotated so that no plant was always nearest the window, and the person measuring did not know which group was which, where that was possible.

5.3 Ethics and Safety

Every investigation involving people, animals or the environment must consider ethics. For studies with human participants, this means informed consent (people agree freely after understanding what is involved), privacy (personal information is kept confidential), and the right to withdraw at any time. For animal studies, it means minimising suffering and using the smallest number of animals that can answer the question. For plant and environmental studies, it means not damaging habitats and

disposing of materials safely.

Safety is part of the method, not an afterthought. Note the risks and the precautions: wearing safety glasses when handling chemicals, washing hands after handling soil, keeping electrical equipment away from water, and following the school or laboratory safety rules. If a risk assessment was completed before the work began, say so. A report that mentions safety shows a careful investigator.

5.4 What Not to Put in Methods

The Methods section should not contain results ("the plants grew well" belongs in Results), opinions ("the experiment was fun" belongs nowhere in a formal report), or background theory (which belongs in the Introduction). Keep it factual and procedural. If a step was changed during the experiment — a lamp broke and was replaced, or watering was missed one day — say so honestly. A reader who knows about the change can judge the results fairly; a reader who is kept in the dark cannot.

Methods checklist

Materials listed with quantities and sizes.
Procedure written step by step, in the past tense.
Number of repeats and group sizes stated.
Independent, dependent and controlled variables named.
How measurements were taken, and how often, explained.
Safety precautions and ethical approval noted.

Chapter 6. Results: Tables, Graphs and Honest Numbers

The Results section presents the findings without explaining them. Interpretation belongs in the Discussion. This separation is important: the reader should be able to see what was found before being told what it means. A Results section that mixes in interpretation forces the reader to accept the writer's opinion before they have seen the evidence.

6.1 Presenting Data in Tables

Tables are best for exact numbers. Every table needs a number and a clear title above it, units in the column headings, and the same number of decimal places throughout a column. Do not put raw data for every single measurement in the main text if there are hundreds of values; give the summary (usually the average) in the main table and put the full raw data in an appendix.

Table 2 below shows a sample results table for the fictional plant-growth experiment used in Chapter 10. It is a model of good practice: the title says what the table shows, the units are in the headings, and the averages are clearly labelled.

Light level	Day 7 (cm)	Day 14 (cm)	Day 21 (cm)	Day 28 (cm)
Low	3.1	5.4	8.2	11.2
Medium	4.0	8.1	13.5	18.3
High	4.8	10.2	17.4	24.6

Table 2. Mean height of bean seedlings (n = 10 per group) grown under three light levels for 28 days. Heights are mean values in centimetres. (Fictional data for illustration.)

After a table, write one or two sentences pointing out the main pattern in words: "Mean height increased in all three groups over 28 days, but the increase was greatest in high light and smallest in low light." Do not repeat every number from the table in the text; the reader can see the numbers. Pick out the pattern, the largest difference, and anything unexpected.

6.2 Presenting Data in Graphs

Graphs are best for showing patterns and trends. A line graph suits measurements taken over time, such as height measured every seven days. A bar chart suits comparisons between separate groups, such as the final heights of three groups. A scatter graph suits the relationship between two measured quantities. Choose the graph type that fits the data, not the one that looks most attractive.

Every graph needs a number and a title, labelled axes with units, and a scale that starts at zero for a bar chart (so that differences are not exaggerated). If more than one group is shown, include a clear key. Never draw a graph by hand and photograph it for a formal report if a computer graph is available; a spreadsheet or graphing program produces a cleaner, more accurate figure.

6.3 Honest Numbers: Averages, Range and Uncertainty

A single measurement can mislead, so results are usually summarised with an average (the mean) and some indication of spread, such as the range (lowest to highest value) or the standard deviation. Report the number of measurements the average is based on — "n = 10" — because an average of ten plants is more trustworthy than an average of two.

Be honest about uncertainty. If the ruler could only be read to the nearest millimetre, do not report heights to two decimal places. If one plant died or a measurement was missed, say so and explain how it was handled. Never invent a missing value, never delete an inconvenient result without saying so, and never round numbers to make a difference look bigger than it is. A small, honest difference is worth more than a large, invented one.

6.4 Writing the Results Text

Write the Results in the past tense and in a logical order: start with the main finding, then the supporting details, then anything unexpected. Use numbers, not vague words: "high-light seedlings were on average 13.4 cm taller than low-light seedlings after 28 days" is useful; "high-light seedlings grew a lot taller" is not. Refer to each table and graph by number in the text — "as shown in Table 2" — so the reader knows where to look.

Chapter 7. Discussion: What the Results Mean — and What They Do Not

The Discussion is where the writer thinks. It is also the section where beginners most often go wrong, either by repeating the Results or by claiming far more than the evidence allows. A good Discussion does four things: it answers the research question, it explains the findings, it admits the limitations, and it suggests what should happen next.

7.1 Answer the Question and Compare With the Hypothesis

Begin by stating the main finding in one or two sentences and saying whether it supports the hypothesis. For example: "The results support the hypothesis. Seedlings in high light grew taller than seedlings in low light at every measurement, and the difference increased over time." If the hypothesis was not supported, say so plainly. A hypothesis that turns out to be wrong is not a failed experiment; it is a genuine discovery, and reporting it honestly is good science.

7.2 Explain the Findings

Use the background science from the Introduction to explain why the results happened. In the plant example, the explanation is that more light allows a higher rate of photosynthesis, so the plant can produce more food for growth — up to the point where light becomes so strong that it damages the leaves. Link each important result to an explanation, and say where the explanation is supported by earlier work and where it is only a suggestion.

7.3 Compare With Earlier Work

If other studies have investigated the same question, compare the findings. Do they agree? If they disagree, why might that be — a different plant, a different light level, a shorter experiment, or a smaller number of repeats? Agreement with earlier work strengthens a conclusion. Disagreement does not necessarily mean the present study is wrong, but it must be discussed rather than ignored.

7.4 Limitations: What the Study Could Not Show

Every study has limitations, and naming them is a strength, not a weakness. Perhaps the experiment lasted only four weeks, so nothing is known about longer-term growth. Perhaps only one type of seed was used, so the results may not apply to other plants. Perhaps the light levels were not measured with a light meter, so "high" and "low" are relative terms. Perhaps the sample was small. State the limitations clearly, and explain how each one might have affected the results. A reader who sees honest limitations trusts the rest of the report more, not less.

7.5 What the Results Do Not Mean

Be careful not to overstate the conclusion. A study of bean seedlings in a classroom does not prove that all plants grow faster in brighter light, and it certainly does not prove anything about plants in a forest or a farm. A correlation — two things changing together — does not prove that one caused the other. And a single experiment rarely settles a question. Use careful language: "suggests," "is consistent with" and "supports" are honest words; "proves" is rarely justified in a school or college study.

Discussion checklist

Main finding stated and linked to the hypothesis.
Each key result explained with background science.
Findings compared with earlier work where possible.
Limitations named honestly, with their likely effect.
No claim made that goes beyond the evidence.

Chapter 8. Conclusion and Recommendations for Further Study

The Conclusion is short — usually one or two paragraphs. It restates the aim, summarises the main finding in a sentence or two, and states the answer to the research question. It should not introduce new results, new references or new arguments. A reader who reads only the Conclusion should learn the answer to the study's question.

A good conclusion mirrors the Introduction: the Introduction asked a question, the Conclusion answers it. For example: "This study set out to investigate the effect of light intensity on the growth of bean seedlings. Over four weeks, seedlings in high light grew taller than seedlings in medium or low light, and the difference increased over time. The findings support the hypothesis that greater light intensity increases growth in bean seedlings within the range tested."

8.1 Recommendations for Further Study

End with one or two sensible suggestions for future work. These should follow naturally from the limitations identified in the Discussion. If the experiment lasted only four weeks, a longer study is a natural next step. If only one plant type was used, testing other plants would be useful. If light was described only as "low, medium and high," measuring the exact light intensity with a meter would improve the next study. Avoid vague recommendations such as "more research is needed" without saying what research and why.

Recommendations may also include practical applications, where they are justified. If the findings suggest that seedlings grow best in bright but not extreme light, a gardener might be advised to place young plants in a bright spot that is sheltered from the harshest midday sun. Keep such advice modest and tied to the evidence; a classroom study does not justify advice to commercial farmers.

Chapter 9. References and Appendices

9.1 Why References Matter

A reference list gives credit to the people whose work the writer used, and it allows the reader to find those sources and check them. Using another person's words or ideas without acknowledgement is plagiarism, even if the words are changed slightly. Plagiarism is treated seriously at every level of education, and it can lead to a report being failed. When in doubt, give a reference.

9.2 How to Reference

Schools and universities use different referencing styles, and the student should follow the style their teacher asks for. The most widely used styles in science are APA and Harvard, but the principles are the same in every style: give the author's name, the year, the title, and enough publication details for the reader to find the source. Be consistent — do not mix two styles in one report.

In the text, a reference is usually given briefly, for example "(Smith, 2020)" or "Smith (2020) found that..." The full details appear in the References list at the end. List the sources in alphabetical order by the first author's surname. Check every reference carefully: a wrong year or a misspelt name suggests carelessness and makes the source hard to find.

Online sources need the same care as printed ones. Give the author or organisation, the date, the title and the web address, and note the date the page was accessed, because web pages change. Prefer reliable sources — textbooks, university pages, government science agencies and peer-reviewed journals — over anonymous pages, social media posts and websites that exist to sell a product.

9.3 Appendices

An appendix holds material that is useful but would interrupt the main text: the full set of raw data, a copy of a questionnaire, a detailed calculation, or extra tables and graphs. Each appendix is labelled (Appendix A, Appendix B) and given a title, and the main text refers to it where relevant: "The full raw data are given in Appendix A." Do not use an appendix to hide important results — anything the conclusion depends on belongs in the main Results section.

Chapter 10. A Complete Worked Example: Light and Plant Growth (Fictional)

This chapter is a short model report. It is a complete example of the structure described in Chapters 1 to 9, written around a clearly labelled fictional experiment. The data are invented for teaching purposes and must not be quoted as real research. Use it as a template: replace the topic, the method and the numbers with your own, and keep the structure.

10.1 Model Report: Title and Abstract

Title: The Effect of Light Intensity on the Growth of Bean Seedlings Over Four Weeks.

Abstract: This study investigated whether light intensity affects the growth of bean seedlings. Thirty seedlings were grown for four weeks under low, medium or high light, with water, soil and temperature kept the same. Height was measured every seven days. After four weeks, seedlings in high light averaged 24.6 cm, compared with 18.3 cm in medium light and 11.2 cm in low light. The results support the hypothesis that greater light intensity increases growth in bean seedlings within the range tested.

10.2 Model Report: Introduction

Plants make their food by photosynthesis, a process that uses light energy, water and carbon dioxide to produce sugars. The rate of photosynthesis increases with light intensity up to a point, after which other factors limit the rate. Gardeners often observe that seedlings kept in a dark room grow tall, pale and weak, while seedlings in bright light grow shorter and greener. However, the size of the difference, and how it develops over time, is not always measured carefully in a classroom.

The aim of this study was to investigate the effect of light intensity on the growth of bean seedlings over four weeks. The research question was: does light intensity affect the growth of bean seedlings? The hypothesis was that bean seedlings grown in high light would grow taller than seedlings grown in low light over four weeks. The independent variable was light intensity; the dependent variable was seedling height; water, soil, pot size and temperature were controlled.

10.3 Model Report: Methods

Thirty bean seeds of the same variety were planted, one per pot, in pots 9 cm in diameter filled with the same potting soil. The pots were divided into three groups of ten. One group was placed under a lamp 60 cm above the plants (high light), one group 120 cm from the same type of lamp (medium light), and one group in a shaded corner of the same room (low light). Each pot received 50 ml of water every second day. The temperature of the room was kept between 20 and 24 degrees Celsius.

Height was measured every seven days from the soil surface to the tip of the tallest leaf, using a ruler marked in millimetres. The same person took all measurements. Plants were rotated within their group each week so that no plant was always nearest the lamp. Safety precautions included keeping the lamp away from water and washing hands after handling soil.

10.4 Model Report: Results

Mean height increased in all three groups over 28 days, but the increase was greatest in high light and smallest in low light (Table 2, Chapter 6). After four weeks, seedlings in high light averaged 24.6 cm tall, compared with 18.3 cm in medium light and 11.2 cm in low light. The difference between the high-light and low-light groups was 13.4 cm. All ten seedlings in each group survived, and no plant showed signs of leaf damage.

10.5 Model Report: Discussion and Conclusion

The results support the hypothesis. Greater light intensity was associated with greater growth at every measurement, and the gap between the groups widened over time. This is consistent with the background science: more light allows a higher rate of photosynthesis, so the plant can produce more food for growth.

The study had limitations. Light intensity was not measured with a meter, so the groups are described only as low, medium and high. Only one type of seed was used, and the experiment lasted only four weeks. Future work could measure light intensity exactly, test other plant types, and continue the experiment for longer to see whether the advantage of high light continues or whether very bright light eventually damages the leaves.

In conclusion, within the range tested, greater light intensity increased the growth of bean seedlings over four weeks. This model report shows the full IMRaD pattern in miniature: a clear question, a repeatable method, honest numbers, a careful interpretation, and a conclusion that does not claim more than the evidence allows.

Chapter 11. Common Mistakes and a Final Checklist

11.1 Common Mistakes

The same mistakes appear in student reports year after year. Knowing them in advance is the best way to avoid them.

1. Writing the report as a story of what the writer did, rather than as evidence for a reader. "First I did this, then I did that" is a diary, not a report. Organise by IMRaD, not by the order of events in the laboratory.

2. Hiding unexpected results. If a plant died, a measurement was lost or a result disagreed with the hypothesis, report it. Hidden results are discovered sooner or later, and they damage trust in everything else in the report.

3. Confusing a conclusion with a guess. "The plants grew because they were happy" is not a scientific explanation. A conclusion must follow from the data and from established science.

4. Copying text from a textbook or the internet without acknowledgement. Even a single copied paragraph is plagiarism. Read the source, understand it, close it, and write the idea in your own words, with a reference.

5. Presenting numbers without units, or with the wrong units. A height of "24.6" means nothing until the reader knows it is 24.6 centimetres. Check every number, every unit and every decimal place.

6. Using a graph that misleads. A bar chart whose axis does not start at zero, a line joining points that should not be joined, or a graph with no labels all mislead the reader, even if the writer did not intend to deceive.

7. Leaving the writing until the night before the deadline. A rushed report has no time for checking numbers, re-reading for sense, or asking a classmate whether the method is clear. Plan the writing over several days, and leave a final day for checking.

8. Forgetting the reader. Long sentences, undefined technical words and missing explanations all assume the reader already knows what the writer knows. Read the report as a stranger would, and explain anything a stranger could not follow.

11.2 Final Checklist

Use this checklist before submitting any report. Tick each item honestly; if an item cannot be ticked, fix it before the report is handed in.

- Title: specific, honest, and no more than about fifteen words.
- Abstract: one paragraph, 150 to 250 words, covering aim, method, main results and conclusion.
- Introduction: background, gap, aim, research question and hypothesis all present.
- Methods: another student could repeat the work from this section alone.
- Results: tables and graphs numbered, titled and labelled with units; text states the

main patterns.

- Discussion: hypothesis answered, results explained, limitations admitted, no over-claiming.
- Conclusion: short, answers the research question, introduces nothing new.
- References: every source used is listed, in a consistent style, with correct details.
- Numbers: all checked against the raw data, units present, decimal places consistent.
- Language: past tense for what was done and found, present tense for established facts, clear and plain throughout.
- Presentation: headings clear, pages numbered, spelling checked, and the report read through once from start to finish.

Chapter 12. Conclusion

A science research report is a skill that improves with practice. The first report a student writes will feel slow and uncertain; the tenth will feel natural. The structure does not change, and the standard does not change: ask a clear question, design a fair test, record honest numbers, and write them up so that another person can understand, check and repeat the work.

The habits taught in this guide — planning before writing, recording results as they happen, separating what was found from what it means, admitting limitations, and giving credit where it is due — are the habits of professional scientists. They are also habits that serve a student well beyond science: in any field, a clear, honest written record is trusted, while a vague or exaggerated one is not.

Start small, follow the IMRaD structure, use the checklist in Chapter 11, and ask a teacher or classmate to read the draft before it is submitted. Every report is a chance to practise. The goal is not a perfect first report, but a report that is a little clearer and a little more honest than the last one — and a reader who finishes it knowing exactly what was done, what was found, and what it means.

Suggested Reading

The following are general, well-known sources for students who want to read further about scientific writing and research methods. They are listed as broad guidance, not as specific citations for any claim in this guide, and the reader should check that the edition available in their library or school is the most recent one.

- General textbooks on scientific writing and communication in science, available in most school and university libraries.
- University writing-centre guides on laboratory reports and the IMRaD structure, published by university academic skills services.
- Style guides for scientific writing, such as the general guidance published by major

scientific publishers on writing clear abstracts, titles and methods sections.

- Textbooks on research methods in science, covering experimental design, variables, sampling, measurement and the honest presentation of data.
- Guides to referencing and avoiding plagiarism, published by university libraries and academic integrity offices.

A practical next step after reading this guide is to read two or three published research papers in a topic of interest — not to copy them, but to see how experienced writers use the IMRaD structure, present tables and graphs, and discuss their limitations. Noticing what makes a paper easy or hard to follow is one of the fastest ways to improve one's own writing.

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End of report.